Research Writing Sample

Selected and adapted from the MSc dissertation

"When Ratings and Markets Disagree: TTT-Informed Covariate Models for Tennis Match Prediction"

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This standalone sample selects and adapts material from the submitted dissertation. The prose has been shortened and reorganised; the analysis, model choices and numerical results are unchanged. This is not a peer-reviewed publication. The figures redraw dissertation-reported aggregates, and no independent statistical rerun was undertaken in preparing this sample.

1 Research problem and questions

Tennis forecasts can point in different directions: a player may have the stronger ATP ranking while an opponent has performed better recently. Statistical ratings provide another representation of strength by estimating ability from previous results and updating those estimates as matches are observed. A rating is therefore a model-based estimate rather than an official ranking (Glickman and Jones, 2025). The forecasting task concerns both the likely winner and the probability attached to that prediction.

Bookmaker probabilities provide a demanding benchmark in tennis forecasting (Kovalchik, 2016; Wilkens, 2021). Average performance, however, can obscure matches in which a rating model and the market identify different winners. This study examines whether publicly available information can improve TrueSkill Through Time (TTT) forecasts, particularly when they disagree with the bookmaker, and whether uncertainty in TTT skill estimates helps identify these cases.

RQ1. To what extent do logistic regression and neural network models combining TTT outputs with public pre-match covariates improve out-of-sample performance relative to TTT?

RQ2. How frequently do TTT and the bookmaker predict different winners, and how do the covariate models perform within these disagreement matches?

RQ3. How do disagreement frequency and prediction error vary across levels of TTT uncertainty?

The contribution is a chronological forecasting comparison in which the bookmaker remains an external benchmark. Logistic regression and neural networks assess the information recoverable from historical public covariates. Disagreement and uncertainty diagnostics complement overall predictive metrics by examining where rating-based forecasts differ from market assessments.

2 Data and chronological processing

The raw data comprised 16 annual Excel workbooks from tennis-data.co.uk covering ATP men's singles in 2010–2025. They contained tournament context, playing conditions, winner and loser identifiers, rankings, ranking points, scores and bookmaker odds. Files were combined chronologically, with dates, column names and text fields standardised. Only completed matches with valid dates, player information and a usable bookmaker probability entered the analytical sample.

For each available individual bookmaker, decimal odds were converted into implied probabilities and normalised to remove the margin. The benchmark was the mean of these margin-adjusted probabilities across available individual bookmakers. Aggregate average and maximum odds fields were excluded. The resulting market probability was used only for comparison, not as a predictor in any fitted covariate model.

The final sample contained 39,154 completed matches involving 1,091 players. This odds-coverage restriction matters for interpretation: the analysis concerns the matches represented in this sample, rather than every professional tennis match. Set-score fields were matched back to the cleaned records by date, tournament, winner and loser. Total games from earlier matches supplied a workload proxy; current-match scores were not pre-match predictors.

Each match was represented by Players A and B, assigned randomly with fixed seed 123 and retained throughout the analysis. The outcome was one if Player A won and zero if Player B won. Every forecast therefore targeted the same event, Player A's victory. Most continuous predictors were A-minus-B differences; absolute differences were retained where the size of the imbalance, rather than its direction, was relevant.

The periods served separate development and evaluation roles. A random split could allow later information to influence predictions for earlier matches, creating look-ahead bias in a sequential forecasting problem (Bergmeir and Benítez, 2012). The chronological design instead reserved the most recent seasons for validation and testing.

Table 1. Chronological partitions and their roles. Adapted from dissertation section 3.1.3 and Table 3.6.

Partition	Years	Matches	Role
Training	2010–2021	28,890	Initial estimation and preprocessing
Validation	2022	2,539	Specification and hyperparameter selection
Test	2023–2025	7,725	Sequential out-of-sample evaluation

The test set was not repeatedly used to choose features or tune the neural network. Its outcomes became available to later estimation histories only after their dates had passed. This distinction separates retrospective development decisions from the expanding-history forecasting evaluation.

Historical covariates and leakage controls

Ranking and ranking-point variables represented relative strength through log-transformed directional advantages and absolute gaps. Because a lower ATP ranking indicates a stronger player, ranking advantage was log ranking B minus log ranking A. Points advantage was log(points A + 1) minus log(points B + 1). These transformations preserved the meaning of a positive advantage for Player A.

Historical form included career, previous-five and previous-ten win rates. Career and surface win rates were smoothed as (previous wins + 1)/(previous matches + 2). A value of 0.5 was assigned when there was no earlier match history. Log-transformed counts of previous matches and matches on the current surface represented experience. Surface, tournament series and round supplied categorical match context.

Rest and workload variables described days since the previous match, the previous 30 days' match count, whether a player competed the day before, previous-match games and seven-day matches, wins and games. Seven-day win rate was 0.5 when no qualifying match existed. These quantities described observable historical workload rather than directly measuring fatigue.

Head-to-head information recorded earlier direct meetings, their log-transformed count and the difference in head-to-head win rates. Common opponents were players previously faced by both competitors, excluding the current pair. Their features included availability, a log-transformed common-opponent count, win-rate differences and differences in accumulated games against those opponents.

All historical quantities were generated using earlier dates. For a match on date t_i, a rolling window of w days contained only matches meeting the condition below. Excluding the entire current date prevented an earlier same-day result from influencing another match when precise within-day ordering was unavailable.

$$t_i - w \leq t_j < t_i$$

For each date, features were calculated for all matches before player histories were updated. Rest was capped at 60 days, with 60 assigned to players lacking an earlier match. Missing previous-match workload used the training-set median number of games. New-player or missing-history indicators represented absent information where relevant, while seven-day and 30-day counts were zero when no qualifying match existed.

Remaining missing numeric predictors used training-set medians. Continuous predictors were standardised using training means and standard deviations, and those preprocessing values were applied unchanged in validation and testing. Categorical variables were converted into indicators where required. Keeping preprocessing fixed, while later re-estimating model parameters, prevented future periods from redefining the development-stage transformation.

3 Modelling framework

TTT extends skill-rating models by representing changing ability across time (Dangauthier et al., 2007). Its temporal structure links skill estimates across matches and supports historical comparability (Landfried and Mocskos, 2025). In this analysis, each player's latent skill was Gaussian, with an estimated mean and standard deviation. All players entered the sample with the same $N(0, 6^2)$ prior.

For match i, the TTT probability and combined skill uncertainty were defined below. Φ is the standard normal cumulative distribution function, μ denotes estimated skill and σ its standard deviation. The performance-noise parameter was $\beta = 1$, the dynamic parameter $\gamma = 0.03$ and the draw probability zero, reflecting binary match outcomes.

$$p_i^{TTT} = \Phi\left(\left(\mu_{a,i} - \mu_{\beta,i}\right) / \sqrt{\left(\sigma_{a,i}^2 + \sigma_{\beta,i}^2 + 2\beta^2\right)}\right)$$

$$u_i^{TTT} = \sqrt{\left(\sigma_{a,i}^2 + \sigma_{\beta,i}^2\right)}$$

At the beginning of each calendar year, TTT was fitted to matches completed before that year and the historical network was run to convergence. Endpoint skill distributions initialised the target year. Within the year, all matches on a date were predicted before any outcome from that date updated the player states. The 2010 sample began from the common prior. This annual historical refitting is distinct from the daily refitting of the subsequent covariate models.

The principal covariate models used the logit-transformed TTT probability and combined uncertainty. The skill-mean difference was retained descriptively, not entered separately. For logistic regression, z_i^{TTT} denotes the TTT log odds and x_i the additional pre-match predictors. The general specification was:

$$\eta_i = \beta_0 + \beta_1 z_i^{TTT} + \beta_2 u_i^{TTT} + \gamma^T x_i \qquad p_i^{LR} = 1 / \left(1 + \exp\left(-\eta_i\right)\right)$$

The intercept and TTT-probability coefficient allow recalibration; additional coefficients assess predictive information beyond the rating. Eight predefined logistic candidates examined TTT calibration, ranking, form and their full combination, then short-term rest/workload and head-to-head, common-opponent or combined extensions. The calibration-only candidate used TTT log odds without the additional feature groups.

A single-hidden-layer feedforward neural network provided a nonlinear alternative (Goodfellow et al., 2016). Logistic hidden activations and a logistic output mapped the inputs to a win probability. Binary cross-entropy fitting with weight decay and BFGS optimisation was implemented through nnet (Venables and Ripley, 2002). Its four families shared the full short-term feature set and added head-to-head, common-opponent or both relational groups.

Model selection and rolling evaluation

Validation log loss determined the preferred specifications; Brier score and accuracy were complementary measures. Logistic regression compared the eight predictor sets defined before validation. Full covariates with short-term and head-to-head information achieved the lowest validation log loss, 0.6084, and became the primary logistic model.

Neural-network selection occurred within and then across the four feature families. Hidden-layer size and weight decay were varied, with three initialisations for each combination. The lowest-validation-loss configuration was retained within each family, and the four retained networks were compared under the same criterion. The short-term family, without relational extensions, had the lowest validation log loss, 0.6038, and became the primary neural network.

The initial family comparisons used a 300-iteration cap. Nonconvergence in the selected short-term family led to a complete rerun of its 48-configuration validation grid and 876 rolling test fits with a 1,000-iteration cap. All final primary validation and test fits converged in the submitted analysis. The selected primary used three hidden units, weight decay 0.01 and initialisation seed 303. The three extensions retained their initial configurations and 300-iteration cap.

After validation, predictor sets, primary identities, neural-network hyperparameters and training-derived preprocessing were fixed. Logistic coefficients and neural-network weights were not fixed. Before every distinct test date d, all predefined models were re-estimated using the expanding history H_d of completed earlier matches:

$$H_d = \{ j : t_j < d \}$$

The first test-date fit used training and completed validation matches. Later fits also used completed matches from earlier test dates. Every same-date forecast used the same historical information, and the date's outcomes entered the history only after all its predictions had been generated. An outcome could therefore inform a later forecast, but not its own or another same-date prediction.

Test forecasts were retained for all predefined specifications, even though validation had already selected the primaries. A lower test score for an extension could be reported as a sensitivity finding; it did not retrospectively replace the selected primary. TTT and the bookmaker were evaluated on the same test matches, as rating and external benchmarks rather than validation-selected fitted candidates.

This expanding, daily rolling-origin evaluation preserves the distinction between fixed modelling decisions and updated parameter estimates. It uses completed outcomes in the way a sequential forecasting system could use them, without allowing test-period results to redefine the selected information sets or hyperparameters.

Performance metrics and diagnostic definitions

All forecasts were expressed as Player A's win probability p_i and evaluated on identical matches within each comparison population. Log loss was the primary model-selection and evaluation metric because it penalises low probability assigned to the observed result, particularly confident errors. Brier score measures squared probability error. For binary outcome Y_i and N evaluated matches:

$$LogLoss = -(1/N) \sum_{i=1}^{N} [Y_i \log(p_i) + (1 - Y_i) \log(1 - p_i)]$$

$$Brier = (1/N) \sum_{i=1}^{N} (p_i - Y_i)^2$$

Probabilities were clipped to $[10^{-15}, 1 - 10^{-15}]$ before log-loss calculation to avoid undefined logarithms. Lower log loss and Brier score indicate better probability forecasts. Accuracy uses a 0.5 threshold, with Player A predicted when $p_i \geq 0.5$. It does not distinguish the confidence attached to the same predicted winner; probabilities 0.51 and 0.90 produce the same classification. Mean probability assigned to the actual winner was also calculated descriptively, but did not determine selection.

Improvement relative to TTT was summarised as the share of the TTT–bookmaker log-loss gap closed. Let L_TTT, L_Book and L_m be losses for TTT, the bookmaker and covariate model m, evaluated on the same population. The measure was:

$$R_m = \left(L^{TTT} - L_m\right) / \left(L^{TTT} - L^{Book}\right)$$

A value of zero means no improvement on TTT; one means matching bookmaker log loss. Values between them indicate partial recovery, with negative values worse than TTT and values above one better than the bookmaker. This measure aids interpretation and does not replace validation log loss as the selection criterion. It was recalculated within the disagreement subset, whose benchmark gap differs from that of all test matches.

Direction disagreement occurs when TTT and bookmaker probabilities identify opposite winners at the 0.5 threshold. Exactly one benchmark direction is correct in such a binary-outcome match. The diagnostic isolates these matches and recalculates probability metrics on the same subset for each forecast. It does not use the covariate models to define the subset.

Uncertainty-group boundaries were the 25th, 50th and 75th percentiles of the training distribution of combined TTT uncertainty. They remained fixed for validation and testing, with open-ended outer groups. Q1 is lowest uncertainty and Q4 highest; test groups need not have equal sizes. Disagreement rates use every test match in a group as denominator, whereas uncertainty-group log losses below use only disagreement matches.

4 Empirical findings

Across all 7,725 test matches, the bookmaker had the lowest probability error and highest accuracy. Both primary covariate models improved on TTT. The neural network had slightly lower log loss and Brier score than logistic regression, while logistic regression had slightly higher accuracy.

Table 2. Primary forecasts on all test matches, 2023–2025 (N = 7,725). Condensed from dissertation Table 4.3; reported results.

Forecast	Log loss	Brier score	Accuracy
Bookmaker	0.5866	0.2019	68.13%
TTT	0.6576	0.2275	63.70%
Logistic regression (primary)	0.6169	0.2148	65.48%
Neural network (primary)	0.6159	0.2143	64.96%

The primary logistic and neural-network models closed 57.3% and 58.8% of the overall test log-loss gap, as reported in dissertation Figure 4.1. These percentages are retained rather than recomputed from rounded table entries. The common-opponent neural-network extension had the lowest observed fitted-model test log loss, 0.6143, and 61.0% gap closure. It remained an extension because the primary was selected on validation performance.

Performance when the predicted winners differ

TTT and the bookmaker disagreed in 7,145 of 39,154 full-sample matches (18.25%). Testing contained 1,496 disagreement matches among 7,725 (19.37%). Within that subset, the bookmaker direction was correct in 919 matches (61.43%) and TTT in 577 (38.57%). The probabilistic comparison therefore concerns a distinct population, not all test matches.

Table 3. The same primary forecasts within test disagreement matches (N = 1,496). Condensed from dissertation Table 4.7; reported results.

Forecast	Log loss	Brier score	Accuracy
Bookmaker	0.6553	0.2318	61.43%
TTT	0.8239	0.3031	38.57%
Logistic regression (primary)	0.7065	0.2560	52.01%
Neural network (primary)	0.6973	0.2517	51.20%

The primary logistic model closed 69.6% of the within-subset gap and the primary neural network 75.1%. The latter had the lowest log loss and Brier score among the fitted models in these disagreement matches, but the bookmaker remained best. These closures use the disagreement subset's benchmark gap and should not be read as the 57.3% and 58.8% overall closures.

Uncertainty and disagreement frequency

The uncertainty analysis compared the fixed, training-defined groups across the test set. Its first question concerned how often TTT and the bookmaker selected different winners within each group. Each rate uses all test matches assigned to that group, including matches where their forecasts agreed. This differs from the disagreement-only population used for the subsequent log-loss comparison.

Disagreement increased across the four groups: 14.28% in Q1, 18.31% in Q2, 21.70% in Q3 and 26.07% in Q4. The Q1-to-Q4 difference was 11.79 percentage points. Q1 represents the least uncertain TTT skill differences and Q4 the most uncertain; the thresholds were not recalculated to divide the test observations into equally sized groups.

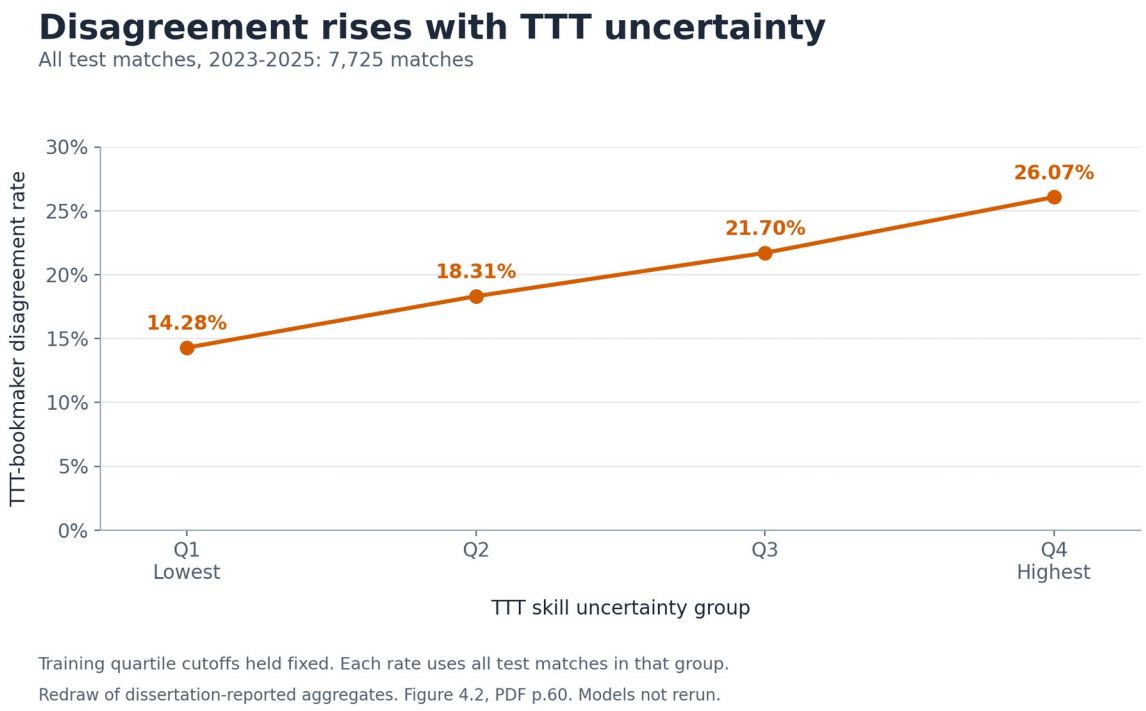


Figure 1. Test disagreement rate across training-defined TTT uncertainty groups. All 7,725 test matches are eligible; each rate's denominator is the full test population in its group. Q1 is lowest uncertainty, Q4 highest. Source-checked redraw from dissertation-reported aggregates, original Figure 4.2 (PDF viewer p.60); no independent statistical rerun.

The increasing disagreement pattern describes the relationship between model uncertainty and forecast direction in this test period. It does not determine which individual forecasts are wrong. Assessing prediction error requires the separate probability comparison within disagreement matches, whose group counts and loss values are considered next.

Uncertainty and error in disagreement matches

The test disagreement subset contained 332 matches in Q1, 396 in Q2, 377 in Q3 and 391 in Q4, totalling 1,496. These are disagreement counts, not the denominators of Figure 1. The bookmaker, TTT and both primary models were compared on the same disagreement observations within each uncertainty group.

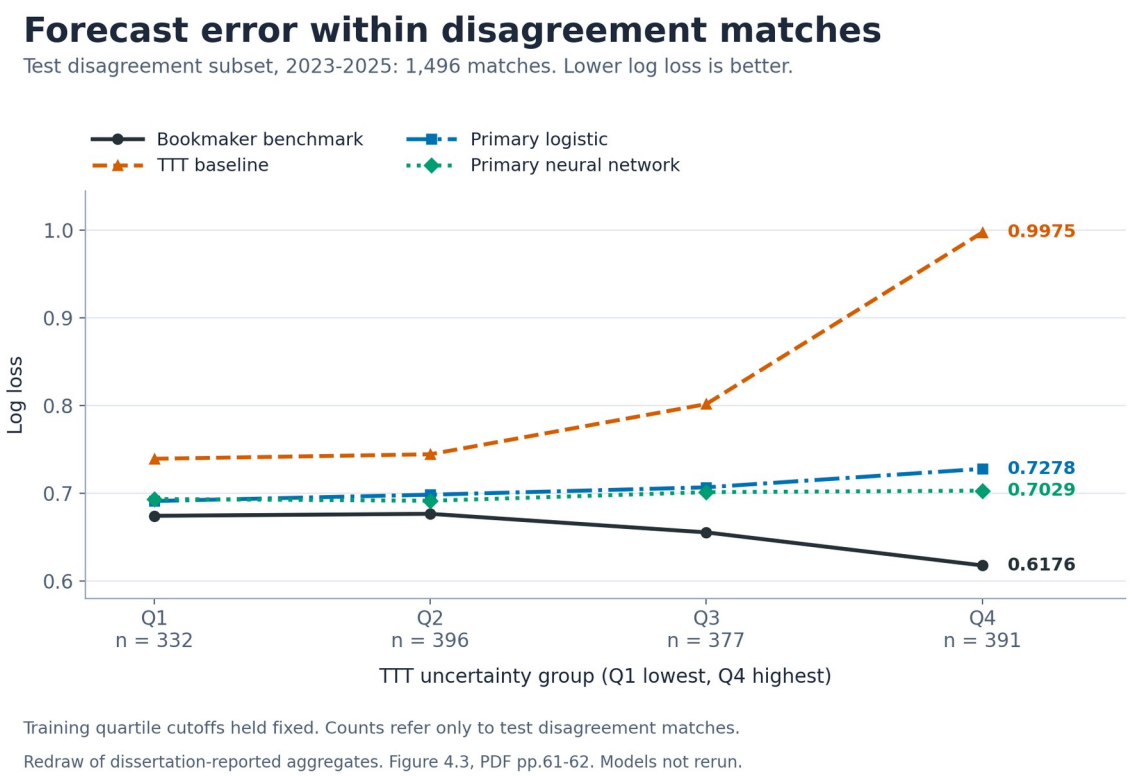


Figure 2. Log loss within test disagreement matches (N = 1,496), grouped by the fixed training uncertainty cutoffs. Subgroup counts are 332, 396, 377 and 391. Lower values indicate better probability forecasts. Source-checked redraw from dissertation-reported aggregates, original Figure 4.3 (PDF viewer pp.61–62); no independent statistical rerun.

TTT log loss was 0.7393 in Q1 and 0.7444 in Q2, increasing to 0.8017 in Q3 and 0.9975 in Q4. Both primaries improved on TTT in every group. Their largest separation from TTT occurred in Q4: logistic log loss was 0.7278, neural-network loss 0.7029 and bookmaker loss 0.6176. Logistic regression was slightly better than the neural network in Q1; the neural network was lower in Q2–Q4.

The bookmaker did not exhibit the same increasing error pattern, with group losses of 0.6742, 0.6765, 0.6553 and 0.6176. The interpretation of these differences is diagnostic and conditional on the selected disagreement subset; no causal effect of uncertainty is inferred.

5 Discussion

Improvement beyond the rating baseline

The first research question is answered by the consistent improvement of the primary models over TTT on test log loss, Brier score and accuracy. TTT summarises changing player strength from previous outcomes, but that summary does not exhaust the information available before a particular match. Ranking, recent form, surface history, workload and matchup records retained predictive value after the rating forecast had been included. TTT's temporal structure remains useful as a foundation for this broader prediction model (Landfried and Mocskos, 2025).

The similar primary-model results qualify the value of additional complexity. A neural network permits nonlinear relationships and interactions, while logistic regression is additive on the log-odds scale (Goodfellow et al., 2016; Hastie et al., 2009). Here, their probabilistic differences were small and the accuracy comparison favoured logistic regression. Under the selected specifications, this suggests that the covariate information contributed more than increased flexibility alone. It is consistent with the flexibility–interpretability trade-off discussed by O'Neill et al. (2021), rather than a general demonstration that more complex models are superior.

The extension findings also require restraint. The common-opponent neural network had the lowest observed overall test loss, but it was not validation-selected and retained the lower iteration cap. Relational extensions did not consistently dominate across validation, all test matches and the disagreement subset. A favourable test result for one extension therefore does not justify redefining the primary or establishing a stable advantage.

Disagreement and uncertainty as diagnostics

The second research question shows that divergent forecast directions occur frequently enough to warrant separate analysis. The bookmaker was correct more often in disagreement matches, while the covariate models recovered a greater fraction of that subset's TTT–bookmaker gap. This indicates that public pre-match information was particularly useful where the general rating differed from the market. It does not identify which information or pricing process produced the remaining bookmaker advantage.

Keeping bookmaker probabilities outside the fitted models makes the comparison interpretable as the contribution of historical public information. Including the market directly could improve forecasting, but would obscure that question (Wilkens, 2021). Even after substantial disagreement-subset improvement, the fitted models did not completely recover the bookmaker advantage.

The third question identifies TTT uncertainty as a useful diagnostic of model limitations. Rating uncertainty concerns the reliability of estimated strength (Glickman, 1999); it is not a complete measure of real-world match uncertainty. Higher values were associated with more disagreement and greater TTT error, while the primary covariate models were less affected. Some uncertain TTT forecasts were nevertheless correct. The evidence supports caution and additional information, not a guarantee of error or a causal explanation.

Taken together, the diagnostics reveal differences that an overall score can conceal. The practical implication is to evaluate both average probabilistic performance and the cases in which a rating disagrees with a stronger benchmark. The present results support a TTT-informed hybrid framework while retaining the bookmaker as an external comparison.

6 Limitations and conclusion

The sample restriction limits generalisation. Only completed ATP men's singles matches with usable bookmaker probabilities were retained. Matches without odds and other levels of professional competition may differ from the analysed population. The observed comparisons should therefore be interpreted within this coverage, rather than applied to all tennis forecasting settings.

The public historical data omitted direct information on injury, physical condition, travel, motivation and late withdrawals. Markets may respond to such information (Wilkens, 2021), but the analysis cannot identify all sources of their advantage. Bookmakers' internal forecasting and pricing procedures are unobserved, so the residual gap cannot be attributed to a particular modelling technique, information source or subsequent price adjustment.

Workload variables were indirect measurements. Rest days, recent match counts and games played record prior competition; they do not directly measure fatigue or recovery. Reid and Duffield (2014) describe physical and recovery processes that match records alone cannot fully observe. Consequently, the predictive contribution of these variables should not be interpreted as an estimated physiological effect.

The fitted structures also constrain the conclusions. Logistic regression used an additive log-odds relationship, and the neural network remained a small feedforward model. Flexibility can represent additional relationships but increases overfitting risk (Hastie et al., 2009; Goodfellow et al., 2016). The primary neural network's higher iteration limit resolved its reported nonconvergence; extensions retained the original cap, so small differences involving them remain descriptive.

TTT uncertainty depends on its skill-distribution assumptions, prior, observed history and updating procedure (Landfried and Mocskos, 2025). It does not encompass every source of uncertainty around a match. Improved prediction after adding a feature group does not establish that the corresponding factor caused the outcome, and the uncertainty–error association is not causal. No betting strategy was evaluated, so these probability comparisons do not establish profitability.

Within those limits, the study answers its central question: public pre-match covariates improved the TTT baseline substantially, closing 57.3% and 58.8% of the overall test log-loss gap for the primary logistic and neural-network models. Neither matched the bookmaker. The larger disagreement-subset improvements and the uncertainty-group patterns show where supplementary information was most useful. Similar primary performance suggests that information construction mattered more than flexibility under these specifications.

Future work could broaden pre-match information, improve workload measurement and stabilise relational features through regularisation or partial pooling. Repeated evaluations and formal uncertainty estimates could test the stability of the diagnostic associations. Extension to later seasons, other tennis populations or prospective forecasts recorded before outcomes would examine generalisation. These are research directions, not analyses completed in this study.

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